Supplementary Lecture Notes

Conservation Equations

1. Mass conservation (continuity equation)

Mass conservation in a flow can be interpreted as: if we look at a system that consists of a fixed set of fluid particles, the mass of the system remains constant. i.e. mass cannot be created or destroyed.

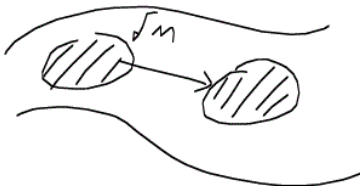


Figure 1

So let's look at a very small set of fluid particles, say with a small amount of mass, δm, and consequently a small volume, δv. The change of the mass of this set of mass therefore must be zero over time:

$$\frac{d(\delta m)}{dt} = 0$$

$$\delta m = \rho \cdot \delta v \tag{1.1}$$

With some manipulations:

$$\frac{d(\delta m)}{dt} = \frac{d(\rho \cdot \delta v)}{dt} = \delta v \frac{d\rho}{dt} + \rho \frac{d(\delta v)}{dt} = 0$$

$$\frac{d\rho}{dt} + \rho \frac{d(\delta V)}{\delta v \cdot dt} = 0 \tag{1.2}$$

As derived in previous lectures, $\frac{d(\delta v)}{\delta v \cdot dt}$ is the flow expansion rate, and

$$\frac{d(\delta v)}{\delta v \cdot dt} = \nabla \cdot \vec{V} \tag{1.3}$$

Therefore we have:

$$\frac{d\rho}{dt} + \rho\nabla \cdot \vec{V} = 0 \tag{1.4}$$

Equ. (4) is the general form of continuity equation in the **Lagrangian system**, in which the observer is following the particle, such that the change of any property of the particle is only a function of time. In contrast, in the **Eulerian system**, the observer sits in the lab coordinate and doesn't follow the flow particle. To give an example, if we ride a boat that flows with the water and watch how the water beside our boat changes (velocity etc), we are in the Lagrangian coordinate. If we sit on the river bank, and watch how water flows at several fixed locations beside us, we are in the Eulerian coordinate.

In Eulerian system, equ. (4) is expressed as:

$$\frac{D\rho}{Dt} + \rho\nabla \cdot \vec{V} = 0 \tag{1.5}$$

where $\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \vec{V} \cdot \nabla$ is called the **material derivative**, which has the same physical meaning as the $\frac{d}{dt}$ operator in the Lagrangian coordinate. Next we shall look at how the definition of the material derivative can be obtained through a simple differential analysis.

2. Material derivative (or total derivative)

It is easier to look at how the properties of a fixed set of mass change with time in Lagrangian system because many of the laws in physics (mass conservation, momentum conservation, energy conservation etc) are stated for a set of fixed mass (or a system). However, in fluid dynamic measurements it is not convenient to track the flow particles and it is much easier to measure the velocity, pressure, temperature etc at fixed locations (i.e. in Eulerian coordinate) because the measuring equipments are usually fixed on the ground.

If we have a measured velocity field $\vec{V} = \vec{V}(x, y, z, t)$, and a measure field of a fluid property, $\phi=\phi(x,y,z,t)$, how do we obtain the rate of change of ϕ of a fixed fluid particle (i.e. $d\phi/dt$ in Lagrangian coordinate) ? Let's look at the fluid particle in the following figure:

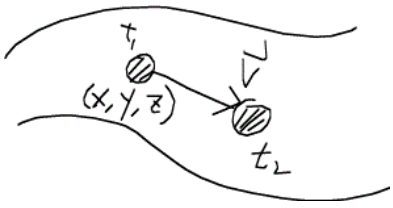


Figure 2

At time t1, the fluid particle is located at (x, y, z), with velocity $\vec{V} = (u, v, w)$, the value of the fluid property is $\phi1=\phi(x,y,z,t1)$. At time $t2 = t1 + \Delta t$, where Δt is very small, the fluid particle

moved to a different location (x', y', z') = (x+uΔt , y+vΔt , z+wΔt), and the fluid property is $\phi 2=\phi(x',y',z',t2)$. Therefore the time change rate of ϕ for the tracked fluid particle in Lagrangian system is:

$$\frac{d\phi}{dt} = \frac{\phi 2 - \phi 1}{\Delta t}|_{\Delta t \to 0} \tag{2.1}$$

Using Taylor expansion:

$$\begin{aligned}
&\phi 2 = \phi(\mathrm{x} + \mathrm{u}\Delta t\,, \mathrm{y} + \mathrm{v}\Delta t\,, \mathrm{z} + \mathrm{w}\Delta t, t1 + \Delta t) + \mathrm{O}(\Delta t^2) \\
&= \phi(\mathrm{x}\,, \mathrm{y}, \mathrm{z}, t1) + \frac{\partial \phi}{\partial \mathrm{t}}\Delta t + \frac{\partial \phi}{\partial \mathrm{x}}(\mathrm{u}\Delta t) + \frac{\partial \phi}{\partial \mathrm{y}}(\mathrm{v}\Delta t) + \frac{\partial \phi}{\partial \mathrm{z}}(\mathrm{w}\Delta t) + \mathrm{O}(\Delta t^2) \\
&= \phi 1 + \Delta t\left(\frac{\partial \phi}{\partial \mathrm{t}} + \mathrm{u}\frac{\partial \phi}{\partial \mathrm{x}} + \mathrm{v}\frac{\partial \phi}{\partial \mathrm{y}} + \mathrm{w}\frac{\partial \phi}{\partial \mathrm{z}}\right) + \mathrm{O}(\Delta t^2) \\
&= \phi 1 + \Delta t\left(\frac{\partial \phi}{\partial \mathrm{t}} + \vec{\mathrm{V}} \cdot \nabla \phi\right) + \mathrm{O}(\Delta t^2)
\end{aligned} \tag{2.2}$$

Therefore we get

$$\frac{d\phi}{dt} = \frac{\phi 2 - \phi 1}{\Delta t}|_{\Delta t \to 0} = \frac{\partial \phi}{\partial \mathrm{t}} + \vec{\mathrm{V}} \cdot \nabla \phi \equiv \frac{D\phi}{Dt} \tag{2.3}$$

Hereby we have proved that the total derivative in Lagrangian coordinate is equivalent to the material derivative in the Eulerian coordinate. The term $\frac{\partial \phi}{\partial \mathrm{t}}$ is called the unsteady term, and $\vec{\mathrm{V}} \cdot \nabla \phi$ is called the convective term, since it is associated with the fluid motion.

3. Species equations

By following the system in Fig. 1, the mass conservation for the ith species can be expressed as:

$$\frac{d(\delta m Y_i)}{dt} = -\oiint_S \vec{n} \cdot (-\rho D_i \nabla Y_i) dA + \delta v M_i \omega_i \tag{3.1}$$

where M_i is the molecular weight of the ith species, and ω_i is the molar production rate of the ith species. The LHS of the equation is the change of the mass for the ith species. The first term on the RHS is the mass of the ith species diffused into the system, and the second term on the RHS is the mass of the ith species produced through chemical reactions.

Since δm is constant,

$$\rho \delta v \frac{dY_i}{dt} = -\oiint_S \vec{n} \cdot (-\rho D_i \nabla Y_i) dA + \delta v M_i \omega_i$$

$$\rho\delta v\frac{dY_i}{dt} = \iiint_V \nabla\cdot(\rho D_i\nabla Y_i)dv + \delta v M_i\omega_i$$

$$\rho\delta v\frac{dY_i}{dt} \approx \delta v\nabla\cdot(\rho D_i\nabla Y_i) + \delta v M_i\omega_i$$

Note that the equal sign holds when $\delta v \to 0$. By removing δv, we have:

$$\rho\frac{dY_i}{dt} = \nabla\cdot(\rho D_i\nabla Y_i) + M_i\omega_i \tag{3.2}$$

In Eulerian system, the above equation is expressed as:

$$\rho\frac{DY_i}{Dt} = \nabla\cdot(\rho D_i\nabla Y_i) + M_i\omega_i \tag{3.3}$$

4. Momentum conservation

Momentum conservation is based on Newton's 2nd Law:

$$\vec{F} = m\vec{a} \tag{4.1}$$

For the example of a small system in Figure 1, equ. (9) becomes

$$\delta m\cdot\vec{a} = \rho\delta v\frac{d\vec{V}}{dt} = \vec{F} \tag{4.2}$$

For a typical fluid flow to be studied in this course, we will mostly consider three types of forces: gravity, hydraulic force (force induced by pressure), and the viscous force:

$$\rho\delta v\frac{d\vec{V}}{dt} = \vec{F} = \rho\delta v\vec{g} + \oiint_S (-p\vec{n})dA + \oiint_S (\vec{n}\cdot\boldsymbol{\tau})dA \tag{4.3}$$

The first term on the RHS is gravity, the second is the hydraulic force acted on the system surface, and the third term is the viscous force on the system surface.

Using the **Divergence Theorem**, surface integration can be converted to volume integration:

$$\begin{aligned}
\rho\delta v\frac{d\vec{V}}{dt} &= \vec{F} \\
&= \rho\delta v\vec{g} + \oiint_S (-p\vec{n})dA + \oiint_S (\vec{n}\cdot\boldsymbol{\tau})dA \\
&= \rho\delta v\vec{g} + \iiint_V (-\nabla p)dv + \iiint_V (-\nabla\cdot\boldsymbol{\tau})dv \\
&\approx \rho\delta v\vec{g} + (-\nabla p)\delta v + (\nabla\cdot\boldsymbol{\tau})\delta v
\end{aligned} \tag{4.4}$$

In the limit $\delta v \to 0$, equ. (12) becomes exact and it can be expressed as:

$$\rho\frac{d\vec{V}}{dt} = \rho\vec{g} - \nabla p + \nabla\cdot\boldsymbol{\tau} \tag{4.5}$$

In Eulerian system, equ. (13) is:

$$\rho\frac{D\vec{V}}{Dt} = \rho\vec{g} - \nabla p + \nabla\cdot\boldsymbol{\tau} \tag{4.6}$$

Which are the **Navier-Stokes equations**.

5. Energy equation

Energy conversion is important to combustion applications. The energy equation is based on the first law, while kinetic energy and various forms of work need to be considered. The basic form of the energy equation is:

$$\frac{d[\delta m(e+\frac{V^2}{2})]}{dt} = -\oiint_S (\vec{n}\cdot\vec{q}\,)dA - \frac{dW}{dt}$$

$$\vec{q} = -\lambda\nabla T + \sum_{i=1,K}[(-\rho D_i\nabla Y_i)e_i] + (\text{Dufour effect, will be neglected}) \tag{5.1}$$

where *e* is the specific internal energy of the mixture, $\vec{q}$ is the energy flux induced by heat transfer and mass diffusion (note that different species has different specific energy), W is the work the done by the system including the work of body force and that by surface forces:

$$-\frac{dW}{dt} = \rho\bar{g}\cdot\vec{V} + \oiint_S [\vec{n}\cdot(\boldsymbol{\tau} - p\boldsymbol{I})]\cdot\vec{V}dA \tag{5.2}$$

Note that the term $\vec{n}\cdot(\boldsymbol{\tau} - p\boldsymbol{I})$ indicates the surface force per unit surface area. Therefore the second term on RHS is the surface work.

Using the Divergence Theorem, (5.1) and (5.2) becomes

$$\frac{d[\delta m(e+\frac{V^2}{2})]}{dt} = -\iiint_V (\nabla\cdot\vec{q})dv + \rho\delta v\bar{g}\cdot\vec{V} + \iiint_V \nabla\cdot[(\boldsymbol{\tau} - p\boldsymbol{I})\cdot\vec{V}]dv$$

$$\begin{aligned}\frac{d\left[\delta m\left(e+\frac{V^2}{2}\right)\right]}{dt} = \quad & -\iiint_V (\nabla\cdot\vec{q})dv + \rho\delta v\bar{g}\cdot\vec{V} - \iiint_V [\vec{\mathrm{V}}\cdot\nabla p + p(\nabla\cdot\vec{\mathrm{V}})]dv \\ & + \iiint_V \left(\vec{\mathrm{V}}\cdot(\nabla\cdot\boldsymbol{\tau}) + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}}\right)dv\end{aligned} \tag{5.3}$$

When $\delta v \to 0$,

$$\begin{aligned}\rho\delta v\frac{d\left(e+\frac{V^2}{2}\right)}{dt} = \quad & -(\nabla\cdot\vec{q})\delta v + \rho\delta v\bar{g}\cdot\vec{V} - \delta v\vec{\mathrm{V}}\cdot\nabla p - p(\nabla\cdot\vec{\mathrm{V}})\delta v \\ & +\delta v\vec{\mathrm{V}}\cdot(\nabla\cdot\boldsymbol{\tau}) + \delta v(\boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}})\end{aligned}$$

or

$$\begin{aligned}\rho\frac{d\left(e+\frac{V^2}{2}\right)}{dt} = \quad & -\nabla\cdot\vec{q} + \rho\bar{g}\cdot\vec{V} - \vec{\mathrm{V}}\cdot\nabla p - p(\nabla\cdot\vec{\mathrm{V}}) \\ & +\vec{\mathrm{V}}\cdot\nabla\cdot\boldsymbol{\tau} + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}}\end{aligned} \tag{5.4}$$

where $-\vec{\mathrm{V}}\cdot\nabla p$ is the **flow work by hydrostatic force**, $p(\nabla\cdot\vec{\mathrm{V}})$ is the **compression work**, $\vec{\mathrm{V}}\cdot(\nabla\cdot\boldsymbol{\tau})$ is the **flow work by viscous force**, and $\boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}}$ is the **dissipation work** (conversion of kinetic energy to internal energy).

To eliminate the kinetic energy term from (5.4), we can dot product velocity to both sides of (4.5)

$$\rho\vec{V}\cdot\frac{d\vec{V}}{dt} = \rho\frac{d\left(\frac{V^2}{2}\right)}{dt} = \rho\bar{g}\cdot\vec{V} - \vec{V}\cdot\nabla p + \vec{V}\cdot(\nabla\cdot\boldsymbol{\tau}) \tag{5.5}$$

Subtract (5.5) from (5.4), then we have

$$\rho\frac{de}{dt} = -\nabla\cdot\vec{q} - p\left(\nabla\cdot\vec{\mathrm{V}}\right) + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}} \tag{5.6}$$

In **Eulerian system**, (5.6) can be written as:

$$\rho\frac{De}{Dt} = -\nabla\cdot\vec{q} - p\left(\nabla\cdot\vec{\mathrm{V}}\right) + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}} \tag{5.7a}$$

For systems under **constant pressure**, enthalpy is typically a preferred variable to use since the RHS of (5.7) can be shortened:

$$\rho\frac{Dh}{Dt} = \rho\frac{D(e+pv)}{Dt} = \rho\frac{De}{Dt} + p\frac{Dv}{vDt} = \rho\frac{De}{Dt} + p\nabla\cdot\vec{\mathrm{V}}$$

i.e.

$$\rho\frac{Dh}{Dt} = -\nabla\cdot\vec{q} + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}} \tag{5.7b}$$

In many combustion systems, the energy equation is typically expressed in terms of temperature. To use temperature as the variable on the LHS of (5.7), the following transformation can be used:

$$\frac{de}{dt} = \frac{d\sum_i(Y_i e_i)}{dt} = \sum_i\left(\frac{dY_i}{dt}e_i + Y_i\frac{de_i}{dt}\right) = \sum_i\left(\frac{dY_i}{dt}e_i\right) + \sum_i\left(Y_i c_{v,i}\,\frac{dT}{dt}\right)$$

$$\frac{de}{dt} = \sum_i\left(\frac{dY_i}{dt}e_i\right) + \frac{dT}{dt}\sum_i\left(Y_i c_{v,i}\,\right) = \sum_i\left(\frac{dY_i}{dt}e_i\right) + c_v\frac{dT}{dt} \tag{5.8}$$

which indicates that the change in energy of a mixture can be caused by both the change in temperature and that in mixture composition. By substituting (3.2) to (5.8), we have

$$\rho\frac{de}{dt} = \sum_i\left(e_i\nabla\cdot(\rho D_i\nabla Y_i) + M_i\omega_i e_i\right) + \rho c_v\frac{dT}{dt}$$

Therefore:

$$\rho c_v\frac{DT}{Dt} = -\sum_i(M_i\omega_i e_i) - \sum_i\nabla\cdot(e_i\rho D_i\nabla Y_i) + \sum_i(\rho D_i\nabla Y_i\cdot\nabla e_i) + \nabla\cdot(\lambda\nabla\mathrm{T}) - p\left(\nabla\cdot\vec{\mathrm{V}}\right) + \boldsymbol{\tau}\colon\nabla\vec{\mathrm{V}}$$

$$\nabla e_i = c_{v,i}\nabla T \tag{5.9}$$

The terms on the RHS are chemical heat release, heat transfer into the fluid particle by mass diffusion and heat conduction, respectively, compression work, and dissipation work, respectively.

In many combustion applications, due to the large amount chemical heat release and significant compression work, the dissipation work can usually be ignored. Furthermore, if assuming the heat capacity, $c_{v,i}$, for all the species are mostly identical, i.e. $c_{v,i} = c_v$,

$$\sum_i (\rho D_i \nabla Y_i \cdot \nabla e_i) \approx c_v \nabla T \cdot \sum_i (\rho D_i \nabla Y_i) = 0$$

Since the net mass flux caused by species diffusion is zero, i.e. $\sum_i (\rho D_i \nabla Y_i) = 0$. In such cases, the energy equation is simplified to:

$$\rho c_v \frac{DT}{Dt} \approx -\sum_i (M_i \omega_i e_i) + \nabla \cdot (\lambda \nabla \mathrm{T} - \sum_i (e_i \rho D_i \nabla Y_i)) - p \nabla \cdot \vec{\mathrm{V}} \tag{5.10}$$

If we further limit the system to spatially homogeneous flows, i.e. $\nabla T = 0$, in a constant volume, eq (5.9) reduces to:

$$\rho c_v \frac{dT}{dt} = -\sum_i (M_i \omega_i e_i) \tag{5.11}$$

Question: in almost all the combustion textbooks, the energy flux in eq. (5.1) is expressed as:

$$\vec{q} = -\lambda \nabla T + \sum_i (-\rho D_i \nabla Y_i) h_i + (\text{Dufour effect})$$

Should we use e_i or h_i in the expression?